

Sum Reduction

PLATFORM

Platform: CodeChef

DIFFICULTY

DIV 3

DATE

Solved: June 3, 2026

Problem

Choose two numbers whose AND is zero.

$$A_i \& A_j = 0$$

Replace them with:

$$A_i + A_j$$

Can array become a single element?

Algorithm

For every number:

For every bit 0..29:

· if $((x \& (1 \ll \text{bit})) \neq 0)$

· $\text{count}[\text{bit}]++$;

If any count > 1: No

Else: Yes

Key Observation

Consider any bit position.

If a bit appears in two different elements:

bit b present in x

bit b present in y

These two elements can never be merged together because:

$$x \& y \neq 0$$

That bit will always remain in at least two elements.

Hence impossible to end with one element.

TC & SC

Time Complexity : $O(30N) \Rightarrow O(N)$

Space Complexity : $O(30) \Rightarrow O(1)$

Example

[1, 2, 4]

001

010

100

Every bit appears once.

Answer = Yes

Condition

For every bit:

$$\text{count}(\text{bit}) \leq 1$$

Sum Reduction

You are given an array A of N elements.

In one operation, you may choose two distinct indices i and j such that the bitwise AND of A_i and A_j is 0, i.e. $A_i \& A_j = 0$.

Then, remove both A_i and A_j from the array, and insert their sum $A_i + A_j$ back into the array.

Determine whether it is possible to reduce the array to exactly one element after performing a finite number of operations.

Input Format

- The first line of input will contain a single integer T , denoting the number of test cases.
- Each test case consists of two lines of input.
 - The first line of each test case contains a single integer N , denoting the size of the array.
 - The second line of each test case contains N space-separated integers $A_1, A_2, \dots, A_N$, denoting the elements of the array.

Output Format

For each test case, output `Yes` if the array can be reduced to exactly one element, otherwise output `No`.

You may print each character of the string in uppercase or lowercase (for example, the strings `yEs`, `yes`, `Yes`, and `YES` will all be treated as identical).

Constraints

- $1 \leq T \leq 100$
- $1 \leq N \leq 100$
- $0 \leq A_i < 2^{30}$

Sample 1:

Input	Output
4 2 3 4 3	Yes No Yes No

Java

```
1 import java.io.*;
2 import java.lang.*;
3 import java.util.*;
4
5 class Codechef
6 {
7     public static void main (String[] args) throws java.lang.Exception
8     {
9         // your code goes here
10        Scanner sc = new Scanner(System.in);
11
12        int T = sc.nextInt();
13
14        while (T-- > 0) {
15            int N = sc.nextInt();
16
17            int[] bitCount = new int[30];
18            boolean possible = true;
19
20            for (int i = 0; i < N; i++) {
21                int x = sc.nextInt();
22
23                for (int b = 0; b < 30; b++) {
24                    if ((x & (1 << b)) != 0) {
25                        bitCount[b]++;
26                        if (bitCount[b] > 1) {
27                            possible = false;
28                        }
29                    }
30                }
31            }
32
33            System.out.println(possible ? "Yes" : "No");
34        }
35    }
36    sc.close();
37 }
38
39 }
```

Test against Custom Input

4
2
3 4
3

Correct Answer

Submission ID: 1285419209

Visualize Code

Run

Submit